



IEEE VR 2024
ORLANDO, FL USA



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Eye Tracking Performance in Mobile Mixed Reality

MOTIVATION

- Eye tracking is an increasingly popular method for interacting with augmented reality (AR) and measuring user attention^{1,2}.
- Evaluating eye tracking across multiple platforms and use cases is challenging due to the lack of standardized metrics.
- Existing calibration methods and accuracy measurements often do not account for walking and scanning in mobile AR settings^{3,4,5}.
- We aim to elucidate differences in eye tracking performance in AR due to device types and stimulus movement.



Figure 1 HoloLens 2 (left), Magic Leap 1 (center), & Meta Quest Pro (right)

USER STUDY TASKS

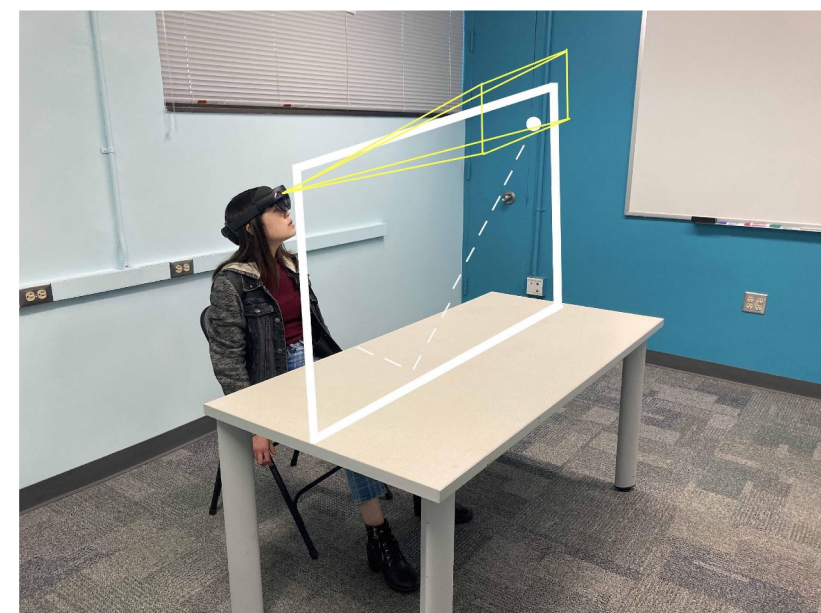
We conducted two separate user studies on Magic Leap (n=36, 1 trial per task) and HoloLens 2 (n=54, 2 trials per task).



Head-Constrained Static (HCS)



Head-Constrained Moving (HCM)



Body-Constrained (BC)



Screen-Stabilized Walking (SSW)



World-Stabilized Walking (WSW)



Hallway (H) (Only HoloLens 2)

Figure 3 Tasks from our user study performed on AR headsets to test the in-built eye-trackers' accuracy

SOLUTION

- Perform eye tracking user studies on the Magic Leap, Meta Quest Pro, and HoloLens 2
- Compare eye tracking performance between the three devices
- Compare eye tracking performance in different AR scenarios (head-constrained, body-constrained, walking)

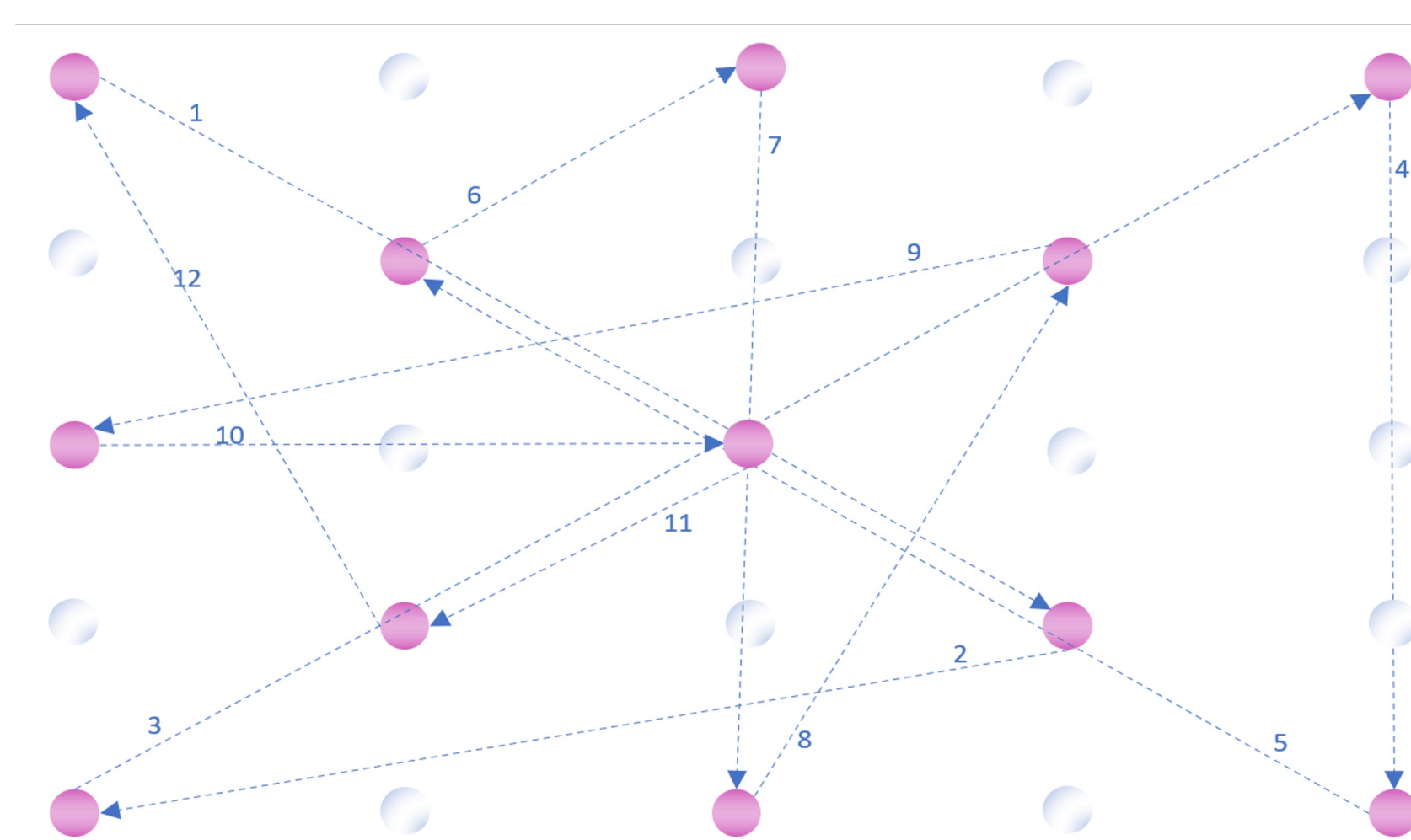


Figure 2 Stimulus moved between a grid of 5x5 on a plane. The paths it took were not obvious or known to the user.

REFERENCES

- [1] B. T. Carter and S. G. Luke. Best practices in eye tracking research. *International Journal of Psychophysiology*, 155:49–62, 2020.
- [2] A. T. Duchowski. *Eye tracking methodology: Theory and Practice*. Springer, 2017
- [3] C. Weigle and D. Banks. Analysis of eye-tracking experiments performed on a Tobii T60. Jan. 2008.
- [4] B. V. Ehinger, K. Groß, I. Ibs, and P. König. A new comprehensive eye-tracking test battery concurrently evaluating the pupil labs glasses and the EyeLink 1000. *PeerJ*, July 2019.
- [5] S. Aziz and O. Komogortsev. An assessment of the eye tracking signal quality captured in the HoloLens 2. In *2022 ACM Symp. on Eye Tracking Research and Applications*. June 2022.

DATA ANALYSIS

Cosine Similarity

The gaze vector v and the target vector u were used to calculate cosine error using the following formula:

$$\theta = \arccos \frac{\vec{u} \cdot \vec{v}}{||u|| ||v||}$$

Spatial Accuracy

To convert the 3D gaze direction and target vectors $v = (v_x, v_y, v_z)$ into 2D screen coordinates, we used the following formulas:

$$x = \arctan\left(\frac{v_x}{v_z}\right), \quad y = \arctan\left(\frac{v_y}{v_z}\right)$$

We then used the 2D screen coordinates to calculate the spatial accuracy:

$$\theta = \sqrt{(x_{target} - x_{gaze})^2 + (y_{target} - y_{gaze})^2}$$

RESULTS: DEVICE COMPARISON

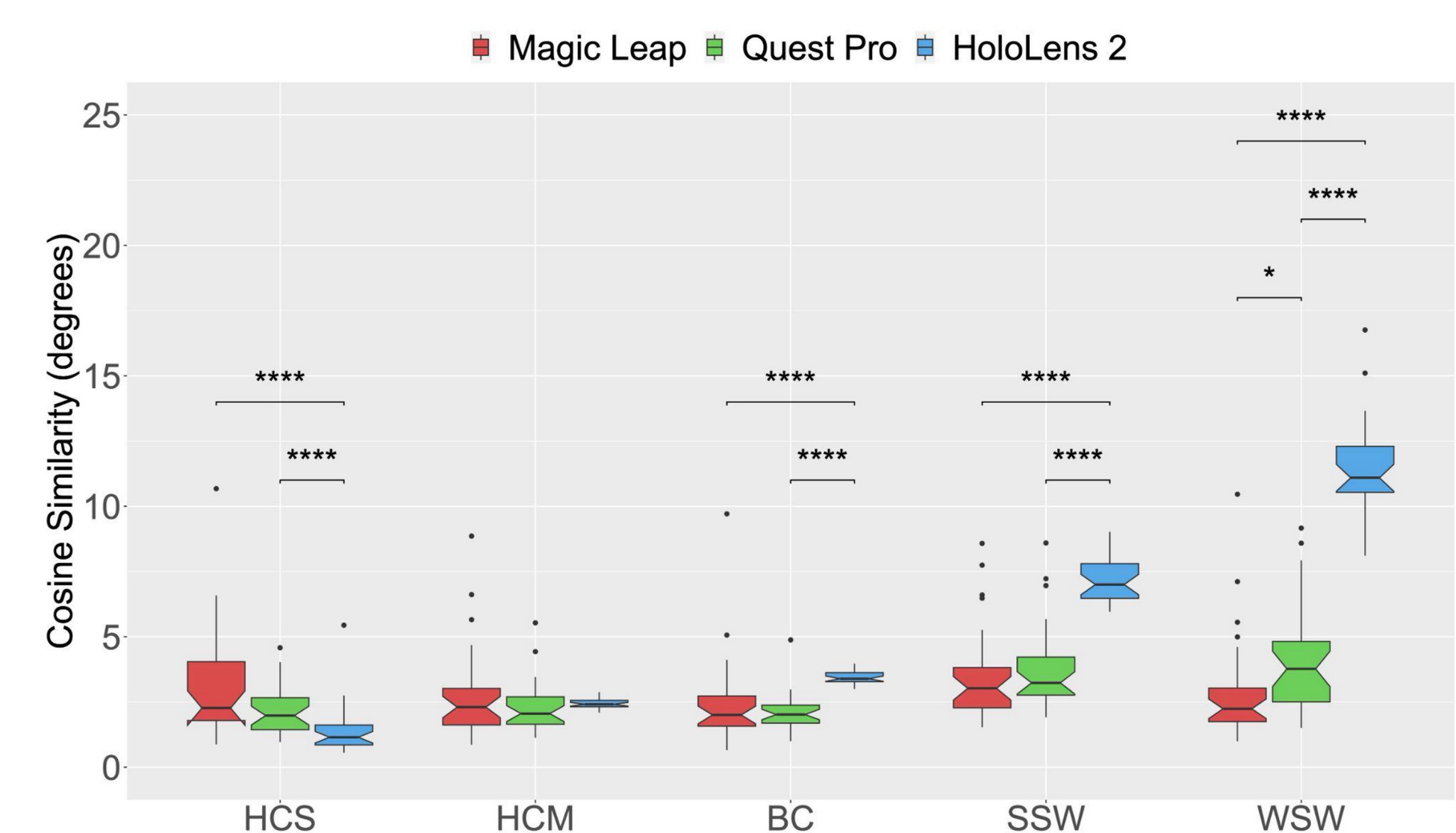


Figure 4 Cosine similarity per task for the Magic Leap, Meta Quest Pro, and HoloLens 2. Notches show 95% confidence interval, * denotes $p < 0.05$, and ** denote $p < 0.01$.

- Magic Leap and Meta Quest Pro data displayed a generally higher accuracy, with more stable fits
- Very limited loss of accuracy due to locomotion or head motion with Magic Leap and Meta Quest Pro
- Wobble in HoloLens 2 headset could be why the it is more accurate only when the head is kept still, compared to the Magic Leap and Meta Quest Pro

The repositories for our eye tracking test suites are at:

- <https://github.com/SatyamA007/MagicLeap-EyeTracking>
- https://github.com/sydneylim/QuestPro_EyeTracking
- <https://github.com/vivianross06/HoloLens-Eye-Tracking>



Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, Tobias Höllerer,
University of California, Santa Barbara



ID: P1323